SUPPORTING ADULT GUIDANCE

Converting spatial units

2D and 3D Shapes. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

The learner converts between spatial units and — crucially — understands **why** the factors differ: length ×10, area ×100, volume ×1000. The scale factor gains an extra power of ten for each dimension, because a square holds 10 × 10 small squares and a cube holds 10 × 10 × 10 small cubes. The work reaches mm, cm and m.

Driving Question: *Why does 1 cm^2 become 100 mm^2 — not 10?*

The mathematics, in plain terms

Along a **length**, 1 cm = 10 mm — a simple ×10. The surprise is **area**: 1 cm^2 is not 10 mm^2 but **100 mm^2**, because a centimetre square is a 10 by 10 grid of millimetre squares (10 × 10 = 100). **Volume** goes further: 1 cm^3 = **1000 mm^3**, a 10 by 10 by 10 stack of millimetre cubes (10 × 10 × 10 = 1000). The pattern is powers of ten, one power per dimension: length 10^1, area 10^2, volume 10^3. The same rule scales up to metres: since 1 m = 100 cm, an area converts by 100^2 = **10 000** (1 m^2 = 10 000 cm^2) and a volume by 100^3 = **1 000 000** (1 m^3 = 1 000 000 cm^3, which is also 1000 litres). To go the other way — smaller units to larger — you divide by the same factor. The single most useful check is: **match the number of tens to the number of dimensions**.

One power of ten per dimension

The heart of this lesson is a single, memorable rule: each dimension adds another factor of ten, so length ×10, area ×100, volume ×1000. It is worth protecting the visual route in first — zooming in to see the 100 little squares and 1000 little cubes, and filling them by hand — because the conversion factors then feel inevitable rather than arbitrary. This connects the whole unit back to place value and powers of ten (the reason multiplying by 10 adds a zero). The commonest error by far is using ×10 for area or volume; the fix is always to picture the grid or the stack. This lesson closes the unit: from describing shapes, to measuring 2D, to building and measuring 3D, and finally to how the units of space themselves scale.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	Zoom & Reveal: zoom into a cm length, square and cube to see 10, 100 and 1000 mm units inside.	~12 min
Movement	Fill It Up: pack a cm^2 with mm^2 squares and a cm^3 with mm^3 cubes, counting to 100 and 1000.	~12 min

Reflection	Spot the Slip: decide whether each unit conversion is right or contains a common mistake.	~9 min
Creativity	Real measures: convert real-world quantities between mm, cm, m and litres.	~5 min
Rest	A pause to let powers of ten settle.	~4 min
Nutrition	Contextual (Mode A): tiling, dosages, land and packaging — where a units slip gets expensive.	~3 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Using ×10 for area (1 cm^2 = 10 mm^2)

Repair: “Draw the cm square as little mm squares — how many rows, and how many in each row? 10 × 10 is …?”

Using ×10 or ×100 for volume instead of ×1000

Repair: “How many layers of 100 fit in the cube? 10 layers of 100 is …?”

Multiplying when converting to a larger unit (should divide)

Repair: “Are you going to smaller units or bigger ones? More small units fit in one big one, so which way — multiply or divide?”

Forgetting to square or cube the factor for m↔cm

Repair: “1 m = 100 cm. For an area, how many times do you use that 100? And for a volume?”

Treating cm^2 and cm^3 as if the units did not matter

Repair: “Is this a length, an area or a volume? The little ^{2} or ^{3} tells you how many tens to use.”

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line

Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson’s Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: multiplying and dividing by 10, 100 and 1000, and the idea of powers of ten. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Keep the picture — and the dimension count — with the learner:

- “Is this a length, an area or a volume?”
- “So how many tens is that — one, two or three?”
- “Picture the little units inside the big one. How many fit — 10, 100 or 1000?”
- “Are we going to smaller units (multiply) or bigger ones (divide)?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

“Why isn't 1 cm^2 just 10 mm^2?”

Because a square has two dimensions. A centimetre square is 10 mm across and 10 mm down, so it holds 10 × 10 = 100 millimetre squares. The Zoom & Reveal panel shows this directly.

“Do they need to memorise 10, 100, 1000, 10 000, 1 000 000?”

No — they need the rule that each dimension adds a factor of ten. From ‘1 m = 100 cm’ a learner can work out the area and volume factors themselves by squaring and cubing.

“How does this connect to the rest of the unit?”

It is the natural finale: having measured length, area and volume, the learner now sees how the units of space themselves scale — a direct application of place value and powers of ten.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Approaching Proficiency	Beginning to engage; names some features and makes a start with support.
Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.
Proficiency	Works through independently and explains why each step is true.

Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.